MD Minhazur Rahman

Data Scientist · Machine Learning Engineer

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Summary

Data scientist and ML engineer with an MSc in Data Science (University of Greenwich) and 3+ years of industry data and IT experience. I ship production-grade machine learning end to end — forecasting, imbalanced classification, streaming, and LLM/RAG — with two models live on GCP Cloud Run (Terraform, keyless CI/CD, monitoring), a Kafka stream scorer proven end-to-end on Kubernetes in CI, and an orchestrated weekly-retraining platform (Dagster, dbt). First-author DOI-indexed preprint. Open to Data Scientist / ML Engineer roles worldwide with relocation and visa sponsorship.

Selected Projects (all public; two live)

Real-Time Demand Forecasting — Production MLOps Pipeline 2026

LightGBM · FastAPI · Docker · Terraform · Cloud Run *github.com/minhazda/synthetic-retail-mlops-pipeline*

- **Cut forecast error 40.8% MAE / 47.3% RMSE vs. a seasonal-naive baseline** by building a typed, tested LightGBM pipeline with a single shared feature module that eliminates training/serving skew.
- **Deployed a live, public inference API on GCP Cloud Run** provisioned entirely as code (Terraform) with keyless GitHub→GCP CI/CD (Workload Identity Federation — no stored secrets).
- **Added production observability** with Prometheus metrics, a Grafana dashboard, and Evidently data-drift reporting on the running service.

Real-Time Card-Fraud Detection — Imbalanced Classification 2026

LightGBM · scikit-learn · FastAPI · Docker · Terraform · Cloud Run *github.com/minhazda/fraud-detection-mlops*

- **Reached ROC-AUC 0.90 and PR-AUC 0.49 (∼8× the base rate) on held-out data** by handling class imbalance (`scale_pos_weight`) and tuning the decision threshold on a separate validation split.
- **Served real-time scoring with built-in explainability** — engineering features from raw transactions at request time and surfacing the risk signals behind each decision in an interactive demo UI.

Event-Driven Fraud Scoring — Kafka Stream Processing 2026

Kafka/Redpanda · Kubernetes · pydantic · Prometheus *github.com/minhazda/streaming-fraud-detection*

- **Built a real-time Kafka/Redpanda stream processor** scoring raw transactions with the fraud model: strict schema validation into a dead-letter queue, an alerts topic, and Prometheus scoring-latency histograms — with zero training/serving skew (a unit test pins stream scores to batch predictions at 10^{-9}).
- **Proved the stack on Kubernetes in CI:** the pipeline unit-tests without a broker via a `MessageBus` protocol, then every push runs it end-to-end against a real Redpanda broker and deploys it onto a kind cluster.

Orchestrated Retraining Platform — Champion/Challenger 2026

Dagster · dbt · DuckDB · LightGBM · GitHub Actions *github.com/minhazda/retail-demand-platform*

- **Orchestrated the full data-to-model graph as Dagster assets:** dbt builds a tested star schema on DuckDB (data-quality tests run as a pipeline stage), then a LightGBM challenger retrains weekly on 1M+ real retail transactions via a scheduled CI job.
- **Gated deployment on evidence, not retraining:** a champion/challenger rule promotes a new model only on a ≥1% holdout-MAE improvement, with champion metrics versioned in git.

Demand Forecasting on Real Retail Data (UCI Online Retail II) 2026

pandas · LightGBM · DuckDB · time-series *github.com/minhazda/real-retail-forecasting*

- **Improved pooled MAE 26.3% over a seasonal-naive baseline (beating it on 47/49 product series)** by documenting defensible cleaning of 1.07M real transactions and reproducing the findings in DuckDB SQL.

Privacy-Preserving RAG Agent — LLM Tool-Calling 2026

LangGraph · ChromaDB · on-device embeddings · FastAPI *github.com/minhazda/privacy-preserving-rag-agent*

- **Shipped a tool-calling RAG agent behind a fail-closed privacy guard** (on-device embeddings, PII redaction) and built a two-tier evaluation: a deterministic CI gate plus a Claude LLM-as-judge harness with Langfuse tracing.

A/B Experiment Analysis Toolkit 2026

statistics · CUPED · sequential testing · Hypothesis *github.com/minhazda/experiment-analysis-toolkit*

- **Built a typed experimentation toolkit** (power analysis, CUPED variance reduction, always-valid sequential testing) whose test suite verifies the *statistical guarantees themselves*: family-wise false-positive rate under continuous peeking and CI coverage over simulation.

Professional Experience

GPH Ispat Limited — Feb 2022 – Aug 2024
Officer — Information Technology Department — *Chittagong, Bangladesh*

- **Automated departmental reporting and operational workflows** by developing internal Python and SQL tools, reducing manual effort across materials-management and reporting processes.
- **Sustained daily operational continuity for enterprise systems** used across departments at one of Bangladesh's largest publicly listed (DSE) steel manufacturers, through proactive maintenance and troubleshooting.
- **Safeguarded data integrity across inventory tracking** by monitoring and reviewing materials-management data feeding company reporting.

Vcube Soft & Tech — 2021 – Feb 2022
QA Engineer — Computer Vision Data — *Chittagong, Bangladesh*

- **Raised training-data quality for production 3D floor-plan reconstruction models** by curating and validating annotated 360° image datasets and enforcing metadata consistency.
- **Built QA validation pipelines** that systematically caught annotation errors before they reached model training.

Education

MSc Data Science — Pass with Merit — Sep 2024 – Jan 2026
University of Greenwich — *London, United Kingdom*

- Dissertation (70%): real-time retail demand forecasting on privacy-preserving synthetic data — 37% MAE reduction; published as a first-author DOI-indexed preprint (Zenodo).
- Modules: Applied Machine Learning 71%, Cloud Computing & HPC 74%, Big Data Analytics, Statistical Methods for Time Series.

BSc Computer Science & Engineering — Jan 2017 – Jan 2021
Independent University, Chittagong — *Chittagong, Bangladesh*

- Final-year project: rare-event detection in surveillance video using 3D-ResNet18 and a custom CNN (PyTorch) on the UCF Crime dataset.

Technical Skills

Languages	Python, SQL, C++, R
ML / DS	scikit-learn, LightGBM, XGBoost, Random Forest, PyTorch; time-series forecasting, imbalanced classification, feature engineering
MLOps / Cloud	Docker, Kubernetes, Terraform (IaC), GCP Cloud Run, GitHub Actions (CI/CD), MLflow, Prometheus, Grafana, Evidently
Streaming / Pipelines	Kafka (Redpanda), Dagster, dbt, DuckDB; schema validation, dead-letter queues, champion/challenger promotion
LLM / GenAI	LangGraph, RAG, ChromaDB, LLM-as-judge evaluation, Langfuse
Serving / Data	FastAPI, pandas, NumPy, SQL; rolling-origin cross-validation, model evaluation
Testing / Quality	pytest, Hypothesis (property-based), mypy, ruff; typed, tested, reproducible repos

Publications, Awards & Research

- **Publication:** Rahman, M. M. (2026). "Synthetic Retail Time-Series for Privacy-Preserving Demand Forecasting." Preprint, Zenodo. DOI: 10.5281/zenodo.19479285.
- **HPC research:** parallel 2D Jacobi solver in C++/OpenMP on an Azure HPC cluster — 4.81× speedup on 8 threads.
- **Awards:** Winner, Game of Presentations 2019 (inter-university); Second Runner-Up, City Bank Bankers' Hunt 2020 (national case competition).
- **Teaching:** supplementary ML instruction to 10+ MSc peers; 4 years of independent STEM/Python tutoring (15+ students).

Reference

Dr. Tuan Vuong — MSc Dissertation Supervisor, University of Greenwich · T.Vuong@greenwich.ac.uk. Further references available on request.